Hierarchical Active Inference with `pymdp`

Discrete High-Level Agent (Policy Selection via $\mathcal{G}$)
+ Continuous Low-Level Agent (Trajectory Tracking)

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1 Overview and Motivation

The previous version of the hierarchical model used a hand-rolled discrete belief update with passive perception only. This document describes the corrected formulation, implemented using the `inferactively-pymdp` library (v0.0.7), which provides:

- Marginal message passing for state inference (replacing softmax heuristic)
- Expected free energy $\mathcal{G}$ for *active* policy selection
- A clean separation of epistemic and pragmatic value in $\mathcal{G}$
- Three candidate vertex-potential proxies, testable against EEG data

Key upgrade from passive to active inference. The previous agent detected surprise but never *decided* anything. The `pymdp` agent selects between two policies at every timestep:

$$\pi \in \{\underbrace{\textsc{Track}}_{0}, \underbrace{\textsc{Interrupt}}_{1}\}$$

Policy selection is driven by expected free energy $\mathcal{G}(\pi)$, which balances epistemic value (resolving uncertainty about the stimulus) against pragmatic cost (disrupting the ongoing trajectory). This provides a principled threshold: weak stimuli do not cross the $\mathcal{G}$ decision boundary and produce *zero* kinematic perturbation; strong stimuli do.

2 Generative Model Matrices

2.1 Likelihood matrix A

$$\mathbf{A} \in [0,1]^{N_o \times N_s}, \qquad A_{o,s} = p(o_t = o \mid s_t = s), \tag{1}$$

with $N_o = 2$ (no-stim / stim) and $N_s = 3$ (TRACK / SURP / RECOVER):

$$\mathbf{A} = \begin{pmatrix} 0.95 & 0.05 & 0.90 \\ 0.05 & 0.95 & 0.10 \end{pmatrix}. \tag{2}$$

SURP is the only state that strongly predicts stimulus presence.

2.2 Transition tensor $\mathbf{B}(\bar{e})$

$$\mathbf{B} \in [0,1]^{N_s \times N_s \times N_\pi}, \qquad B_{s',s,\pi} = p(s_t = s' \mid s_{t-1} = s,\ \pi_t = \pi). \tag{3}$$

Columns sum to 1. The tensor has two slices, one per policy. Both slices are modulated by normalised efference copy $\bar{e}(t) = \|\boldsymbol{a}(t)\| / a_{\max} \in [0,1]$:

$$p(\textsc{Track} \to \textsc{Surp} \mid \pi) = p_0^\pi + \alpha_B\, \bar{e}, \tag{4}$$

$$p(\textsc{Surp} \to \textsc{Rec} \mid \pi) = q_0^\pi + \alpha_B\, \bar{e}, \tag{5}$$

$$p(\textsc{Rec} \to \textsc{Track} \mid \pi) = r_0^\pi + \alpha_B\, \bar{e}. \tag{6}$$

The Interrupt policy has higher base rates $q_0^1 \gg q_0^0$, $r_0^1 \gg r_0^0$: it resolves surprise faster.

Dynamic B in `pymdp`. `pymdp` does not natively support step-varying $\mathbf{B}$. We recompute $\mathbf{B}(\bar{e}(t))$ at every timestep and assign it directly to `agent.B` before calling `agent.infer_policies()`. No agent reinitialisation is required.

2.3 Preference vector C

Log-preferences over observations encode that the agent prefers not to observe surprising stimuli:

$$\mathbf{C} = \big[\ln 0.9,\ \ln 0.1\big]^\top. \tag{7}$$

This enters the *pragmatic* term of $\mathcal{G}$ (Sec. 4).

2.4 Prior over initial states D

$$\mathbf{D} = [0.95,\ 0.025,\ 0.025]^\top. \tag{8}$$

3 State Inference — Marginal Message Passing

At each timestep `pymdp` computes the posterior $q(s_t) \in \Delta^{N_s}$ via marginal message passing:

Step 1 — Prior prediction.

$$\boldsymbol{s}_{\text{prior}} = \mathbf{B}_{:,:,\hat{\pi}}\ \boldsymbol{q}(s_{t-1}), \tag{9}$$

where $\hat{\pi}$ is the policy selected at the previous step.

Step 2 — Likelihood update.

$$\ln q(s_t) = \ln \boldsymbol{s}_{\text{prior}} + \mathbf{A}_{o_t,:}^\top \quad \Rightarrow \quad q(s_t) = \sigma(\ln \boldsymbol{s}_{\text{prior}} + \ln \mathbf{A}_{o_t,:})\,, \tag{10}$$

where $\sigma(\cdot)$ is the softmax.

Surprise.

$$p(o_t) = \mathbf{A}_{o_t,:}\ \boldsymbol{q}(s_t), \qquad \xi_t = -\ln p(o_t). \tag{11}$$

> `pymdp` **call:** `qs = agent.infer_states(obs)`
> Returns `qs[0]`, shape `(NUM_STATES,)`. `obs` is a list of integers, one per modality.

4 Policy Selection — Expected Free Energy

4.1 Definition of $\mathcal{G}$

The expected free energy for policy π over a horizon τ is:

$$\mathcal{G}(\pi) = \sum_\tau \mathbb{E}_{q(o_\tau, s_\tau|\pi)}\Big[\underbrace{\ln q(s_\tau|\pi) - \ln q(s_\tau|o_\tau,\pi)}_{\text{epistemic value}} + \underbrace{\ln q(o_\tau|\pi) - \ln p(o_\tau)}_{\text{pragmatic value}}\Big]. \tag{12}$$

For one-step lookahead ($\tau = 1$) this reduces to:

$$\mathcal{G}(\pi) = \underbrace{-\mathbb{E}_{q(s|\pi)}\big[H[\mathbf{A}_{:,s}]\big]}_{\text{epistemic: info gain}} - \underbrace{\mathbb{E}_{q(o|\pi)}\big[\ln p(o)\big]}_{\text{pragmatic: preferences}}\ , \tag{13}$$

where $H[\mathbf{A}_{:,s}] = -\sum_o A_{o,s} \ln A_{o,s}$ is the entropy of the likelihood column for state s, and $\ln p(o) = \mathbf{C}$.

Interpretation for the two-policy model.

- **Epistemic term:** INTERRUPT has lower expected likelihood entropy because it routes the agent through RECOVER (a more predictable state) — it *resolves* uncertainty.
- **Pragmatic term:** INTERRUPT incurs a pragmatic cost because entering SURP risks observing the unpreferred stimulus. But this is offset by the epistemic gain.
- **Threshold:** $\mathcal{G}(\text{INTERRUPT}) < \mathcal{G}(\text{TRACK})$ only when surprise ξ_t is large enough — this is the free threshold mechanism that produces all-or-nothing perturbations.

4.2 Policy posterior

$$q(\pi) = \sigma(-\gamma_G\, \mathcal{G}(\pi))\,, \qquad q(\pi) \in \Delta^{N_\pi}, \tag{14}$$

where $\gamma_G > 0$ is the policy precision (temperature). High γ_G makes policy selection more deterministic.

`pymdp` calls:
`q_pi, G = agent.infer_policies()`
Returns `q_pi` shape `(NUM_POLICIES,)` and `G` shape `(NUM_POLICIES,)`.

`action = agent.sample_action()`
Returns action index (0 or 1) drawn from `q_pi`.

5 Precision Coupling

5.1 Surprise pulse $\gamma(t)$ — gated by policy

The critical upgrade from the previous version: γ only accumulates when the INTERRUPT policy is selected.

$$\dot{\gamma} = -\frac{\gamma}{\tau_\gamma} + \lambda_\gamma\, \xi_t\, \underbrace{\mathbb{1}[\pi_t = \text{INTERRUPT}]}_{\text{policy gate}}, \qquad \gamma \geq 0. \tag{15}$$

This makes the kinematic perturbation *conditional on policy selection*: stimuli that do not cross the $\mathcal{G}$ threshold produce zero kinematic effect regardless of their physical intensity.

5.2 Low-level sensory precision

$$\mathbf{\Pi}_s^{\text{low}}(t) = \frac{1}{1+\gamma(t)}\, \mathbf{\Pi}_s^{\text{base}}, \tag{16}$$

z-entries always zero (unobserved axis).

5.3 High-level sensory precision

$$\Pi_s^{\text{high}}(t) = \frac{\Pi_s^{\text{high,base}}\,(1+\gamma(t))}{1+\alpha_{\text{eff}}\,\bar{e}(t)}. \tag{17}$$

Independent of $\mathbf{\Pi}_s^{\text{low}}$ — no conservation law.

6 Low-Level Continuous Agent

Unchanged from the single-agent formulation except $\mathbf{\Pi}_s^{\text{low}}(t)$ is now time-varying via Eq. (16).

6.1 Belief update

$$\dot{\boldsymbol{\mu}} = \boldsymbol{f}_{\text{model}}(\boldsymbol{\mu}, \boldsymbol{a}) + \kappa_\mu \big(\boldsymbol{\Pi}_s^{\text{low}}(t)\, \boldsymbol{\varepsilon}_s - \boldsymbol{\Pi}_p\, \boldsymbol{\varepsilon}_p\big). \tag{18}$$

6.2 Precision-modulated PD action

$$u_1 = \frac{m_1}{1+\gamma}\big[K_P(p_1^* - \mu_{p_1}) + K_D(v_1^* - \mu_{v_1})\big], \tag{19}$$

$$u_2 = \frac{m_2}{1+\gamma}\big[K_P(p_2^* - \mu_{p_2}) + K_D(v_2^* - \mu_{v_2})\big] + m_2 g. \tag{20}$$

7 Vertex Potential Proxies

Three candidates are stored, all testable against empirical EEG amplitudes:

Symbol	**Expression**	**Neural interpretation**
$\text{VP}_{\Pi\varepsilon}$	$\Pi_s^{\text{high}}(t) \cdot \lVert \boldsymbol{\varepsilon}_{s,xy} \rVert$	Precision-weighted prediction error; attenuated by efference via Π_s^{high} denominator
$\text{VP}_{\mathcal{G}}$	$-\min_\pi \mathcal{G}(\pi)$	Negative minimum EFE; peaks when one policy is strongly preferred
$\text{VP}_{q\pi}$	$q(\pi = \textsc{Interrupt})$	Posterior probability of interruption; simplest proxy, directly interpretable

Predicted anti-correlation. $\bar{e}(t)$ large (movement) suppresses Π_s^{high} via its denominator $\Rightarrow$ $\text{VP}_{\Pi\varepsilon}$ smaller. Simultaneously, $\bar{e}$ increases $p(\textsc{Track} \to \textsc{Surp})$ via $\mathbf{B}$, so a stimulus that crosses the $\mathcal{G}$ threshold does so with larger ξ and drives a larger γ pulse $\Rightarrow$ bigger kinematic disruption. The two effects are independent, share one cause ($\bar{e}$), and produce opposite signs in the two observables.

8 Full Algorithm

Algorithm 1 Hierarchical AIF — `pymdp` High-Level Agent

Require: Parameters; noise arrays; `pymdp` Agent initialised with $\mathbf{A}, \mathbf{B}_0, \mathbf{C}, \mathbf{D}$

Init: $\boldsymbol{x}_{\text{real}}, \boldsymbol{\mu} \leftarrow \boldsymbol{x}_0$; $\gamma \leftarrow 0$; $\boldsymbol{a}_{\text{prev}} \leftarrow \mathbf{0}$; $\pi \leftarrow \textsc{Track}$
for $k = 0, \ldots, N-1$ **do**
  $t \leftarrow k\Delta t$; $\boldsymbol{x}^* \leftarrow \textsc{DesiredState}(t)$; $o_t \leftarrow \mathbb{1}[\text{stimulus}]$

  — **Efference copy** —
  $\bar{e} \leftarrow \|\boldsymbol{a}_{\text{prev}}\| / a_{\max}$

  — **High-level: update B** —
  `agent.B` $\leftarrow \mathbf{B}(\bar{e})$ *pymdp: direct assignment*

  — **High-level: state inference** —
  `qs` $\leftarrow$ `agent.infer_states([`o_t`])` *pymdp: marginal message passing*
  $q(s_t) \leftarrow$ `qs[0]`; $\xi \leftarrow -\ln(\mathbf{A}_{o_t,:}\, q(s_t))$

  — **High-level: policy selection** —
  `q_pi, G` $\leftarrow$ `agent.infer_policies()` *pymdp: expected free energy*
  $\pi \leftarrow$ `agent.sample_action()[0]` *pymdp: softmax draw*

  — **Precision pulse (policy-gated)** —
  $\gamma \leftarrow \gamma + \big(-\gamma/\tau_\gamma + \lambda_\gamma\, \xi\, \mathbb{1}[\pi{=}\textsc{Interrupt}]\big)\Delta t$

  — **Low-level precision** —
  $\sigma \leftarrow 1/(1+\gamma)$; $\boldsymbol{\Pi}_s^{\text{low}} \leftarrow \sigma\, \boldsymbol{\Pi}_s^{\text{base}}$; $\Pi_s^{\text{high}} \leftarrow \Pi_s^{\text{h,base}}(1+\gamma)/(1+\alpha_{\text{eff}}\bar{e})$

  — **Low-level: observe, errors, free energy** —
  $\boldsymbol{y}_{\text{obs}} \leftarrow$ observe x-y only
  $\boldsymbol{\varepsilon}_p \leftarrow \boldsymbol{\mu} - \boldsymbol{x}^*$; $\boldsymbol{\varepsilon}_s \leftarrow \boldsymbol{y}_{\text{obs}} - \boldsymbol{\mu}$
  $\mathcal{F}_{\text{low}} \leftarrow \frac{1}{2}(\boldsymbol{\varepsilon}_p^\top \boldsymbol{\Pi}_p \boldsymbol{\varepsilon}_p + \boldsymbol{\varepsilon}_s^\top \boldsymbol{\Pi}_s^{\text{low}} \boldsymbol{\varepsilon}_s)$

  — **Low-level: action** —
  $u_1 \leftarrow \sigma\, m_1[K_P(\ldots) + K_D(\ldots)]$; $u_2 \leftarrow \sigma\, m_2[\ldots] + m_2 g$; clip; $\boldsymbol{a}_{\text{prev}} \leftarrow \boldsymbol{a}$

  — **World + belief update** —
  $\boldsymbol{x}_{\text{real}} \leftarrow \boldsymbol{x}_{\text{real}} + \boldsymbol{f}_{\text{proc}}(\ldots)\Delta t$; $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + [\boldsymbol{f}_{\text{model}} + \kappa_\mu(\boldsymbol{\Pi}_s^{\text{low}} \boldsymbol{\varepsilon}_s - \boldsymbol{\Pi}_p \boldsymbol{\varepsilon}_p)]\Delta t$

  — **Store VP proxies** —
  $\text{VP}_{\Pi\varepsilon} \leftarrow \Pi_s^{\text{high}}\, \|\boldsymbol{\varepsilon}_{s,xy}\|$; $\text{VP}_{\mathcal{G}} \leftarrow -\min_\pi G(\pi)$; $\text{VP}_{q\pi} \leftarrow q(\pi{=}\textsc{Interrupt})$
end for

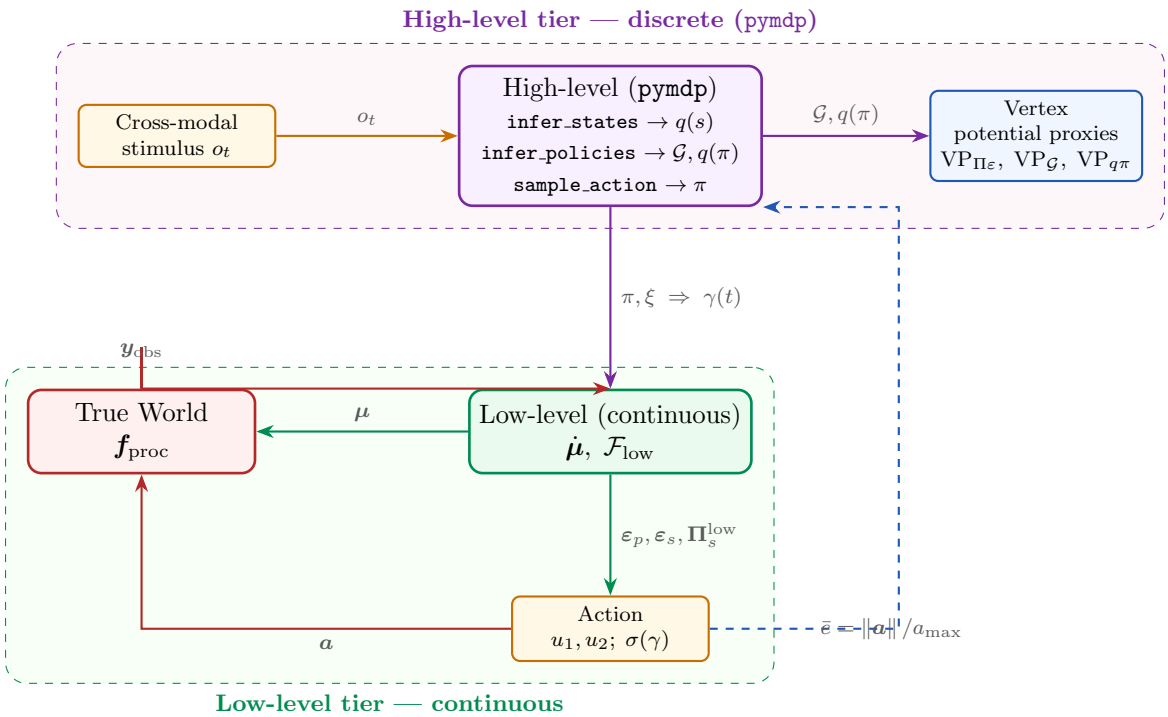

Figure 1: Two-tier hierarchical AIF with `pymdp` high-level agent. Blue dashed: efference copy. Purple: `pymdp` inference/policy pipeline. Orange: cross-modal stimulus and precision pulse.

9 System Diagram

10 Parameter Summary

11 Model Comparison Strategy

The three vertex-potential proxies differ in their neural interpretability and their dependence on model components:

Q1. Which VP proxy best fits empirical N1/P3 amplitudes?
Rank $\mathrm{VP}_{\Pi\varepsilon}$, $\mathrm{VP}_{\mathcal{G}}$, $\mathrm{VP}_{q\pi}$ by R^2 against EEG at stimulus onset. $\mathrm{VP}_{q\pi}$ is most interpretable; $\mathrm{VP}_{\mathcal{G}}$ is most theoretically grounded.

Q2. Can α_B and α_{eff} be jointly fitted?
These two parameters control the movement/stationary ratio in kinematic disruption and VP attenuation respectively. A grid search over $(\alpha_B, \alpha_{\mathrm{eff}})$ should recover a ridge of solutions consistent with the empirical ratio.

Q3. Is γ_G identifiable from kinematics alone?
High $\gamma_G \Rightarrow$ deterministic INTERRUPT selection $\Rightarrow$ step-like perturbation onset. Low γ_G $\Rightarrow$ graded onset. The sharpness of the triphasic onset constrains γ_G.

Q4. Does the policy gate (Eq. 15) improve fit?
Compare against the ungated version (previous model) to test whether the threshold mechanism is necessary to reproduce all-or-nothing perturbations.

Table 1: All parameters. Low-level parameters unchanged from v1.

Symbol	**Description**	**Default**	**Role**
pymdp agent			
γ_G	Policy precision (temperature on $\mathcal{G}$)	16.0	Determinism of policy selection
α_B	Efference $\rightarrow$ B modulation	2.0	Destabilisation during movement
Precision coupling			
τ_γ	Surprise pulse decay time	0.3 s	Perturbation duration
λ_γ	Surprise $\rightarrow \gamma$ gain	1.5	Perturbation amplitude
α_{eff}	Efference $\rightarrow \Pi_s^{\text{high}}$ suppression	0.05	VP attenuation during movement
$\Pi_s^{\text{h,base}}$	Baseline high-level precision	10.0	VP baseline
B matrix base rates			
p_0^0, p_0^1	Base TRACK$\rightarrow$SURP (TRACK / INT policy)	0.05, 0.10	Spontaneous surprisability
q_0^0, q_0^1	Base SURP$\rightarrow$REC	0.25, 0.70	Recovery speed
r_0^0, r_0^1	Base REC$\rightarrow$TRACK	0.20, 0.50	Re-engagement speed